2 A student adds excess large pieces of magnesium carbonate, $MgCO_3$, to dilute hydrochloric acid, HCl, and measures the volume of carbon dioxide gas, CO_2, given off.

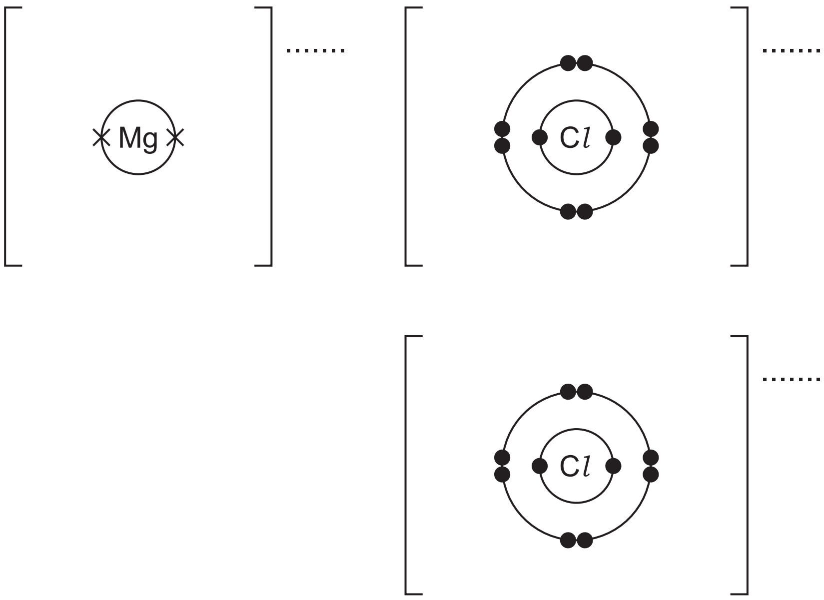
(a) Add the missing state symbols to the chemical equation for the reaction.

$$MgCO_3 \ldots\ldots + 2HCl\ldots\ldots \rightarrow MgCl_2(aq) + H_2O \ldots\ldots + CO_2 \ldots\ldots$$ [1]

(b) Complete the dot-and-cross diagram to show the electron arrangement of the ions in magnesium chloride.

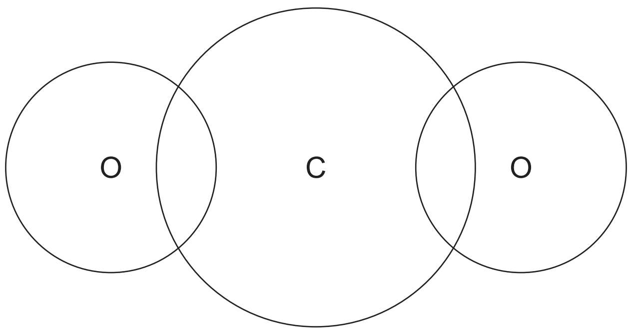
The inner shells have been drawn.

Give the charges on the ions.



[3]

(c) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of carbon dioxide.
Show outer shell electrons only.



[2]

2 **(a)** Atoms are made of protons, neutrons and electrons. Atoms of the same element are known as isotopes.

(i) Complete the table.

particle	relative charge	relative mass
electron		$\frac{1}{1840}$
neutron		
proton	+1	

[2]

(ii) $^{24}_{12}Mg$ and $^{25}_{12}Mg$ are isotopes of magnesium.

Complete the table to show the numbers of electrons, neutrons and protons in these isotopes of magnesium.

isotope	number of electrons	number of neutrons	number of protons
$^{24}_{12}Mg$			
$^{25}_{12}Mg$			

[2]

(iii) Explain why magnesium ions have a charge of 2+.

....

.... [1]

(b) Mg^{2+} ions have the electronic structure 2,8.

Give the formula of the following particles which have the same electronic structure as Mg^{2+} ions.

- a cation (positive ion)

....

- an anion (negative ion)

....

- an atom

....

[3]

[Total: 8]

3 This question is about sodium and compounds of sodium.

(a) **(i)** Describe the bonding in a metallic element such as sodium.

You may include a diagram as part of your answer.

....

....

.... [3]

(ii) Describe how solid sodium conducts electricity.

.... [1]

(b) Some properties of sodium chloride are shown:

- melting point of 801 °C
- non-conductor of electricity when solid
- conductor of electricity when molten
- soluble in water.

(i) Name the type of bonding in sodium chloride.

.... [1]

(ii) Explain why sodium chloride conducts electricity when molten.

....

.... [1]

3 Transition elements are found in the middle block of the Periodic Table.

(a) Chromium has several isotopes. Manganese has only one isotope.

(i) State what is meant by the term *isotopes*.

....

.... [2]

(ii) State the nucleon number of manganese.

.... [1]

(iii) Complete the table to show the number of protons, neutrons and electrons in a $^{52}_{24}Cr^{3+}$ ion.

protons	neutrons	electrons

[3]

(b) One chemical property of transition elements is that they form coloured compounds.

(i) Give the colours of the following hydrated salts.

- hydrated copper(II) sulfate ....
- hydrated cobalt(II) chloride ....

[2]

(ii) State two **other** chemical properties of transition elements.

1 ....

2 ....

[2]

(c) Transition elements and Group I elements are metals. They share many physical properties including the ability to:
- conduct electricity
- be hammered into shape.

(i) Explain why transition elements and Group I elements conduct electricity.

.... [1]

(ii) State the property that describes a material which can be hammered into shape.

.... [1]

2 **(a)** $^{32}_{16}S$ and $^{33}_{16}S$ are isotopes of sulfur.

Use your knowledge of protons, neutrons and electrons to answer the following questions.

(i) Describe how these isotopes of sulfur are the same and how they are different.

same ....

....

different ....

....
[3]

(ii) Explain why each of these isotopes have an overall charge of zero.

....

.... [1]

(iii) Explain why both isotopes have the same chemical properties.

....

.... [1]

(b) Sulfide ions, S^{2-}, have the electronic structure 2,8,8.

(i) Explain why sulfide ions have a charge of 2–.

....

.... [1]

(ii) Give the formula of:

- an anion which has the same electronic structure as S^{2-}

....

- a cation which has the same electronic structure as S^{2-}.

....
[2]

[Total: 8]

1 The table shows the numbers of protons, neutrons and electrons in particles **A** to **I**.

particle	protons	neutrons	electrons
A	1	0	0
B	6	6	6
C	6	8	6
D	10	10	10
E	16	16	18
F	17	18	17
G	18	22	18
H	19	20	19
I	20	20	18

Answer the following questions about particles **A** to **I**. Each letter may be used once, more than once or not at all.

(a) State which of the particles **A** to **I**:

(i) is an anion .... [1]

(ii) are cations .... and .... [2]

(iii) are noble gas atoms .... and .... [2]

(iv) is a halogen atom .... [1]

(v) is a Group I atom .... [1]

(vi) have the same nucleon number .... and .... [1]

(vii) causes acidity in aqueous solutions .... [1]

(viii) is used to define the relative atomic mass of elements. .... [1]

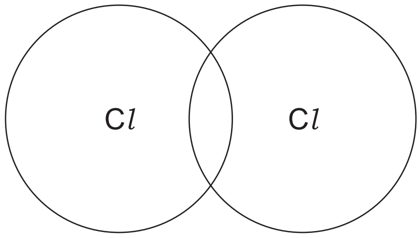
(b) Explain why **B** and **C** are isotopes of the same element.

....

.... [2]

[Total: 12]

(e) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of chlorine, Cl_2.
Show the outer electrons only.



[1]

(f) The melting points and boiling points of chlorine and potassium chloride are shown.

	melting point / °C	boiling point / °C
chlorine	–101	–35
potassium chloride	770	1500

(i) Deduce the physical state of chlorine at –75 °C. Use the data in the table to explain your answer.

physical state ....

explanation ....

....
[2]

(ii) Explain, in terms of structure and bonding, why potassium chloride has a much higher melting point than chlorine.

Your answer should refer to the:
- types of particle held together by the forces of attraction
- types of forces of attraction between particles
- relative strength of the forces of attraction.

....

....

....

....

.... [3]

[Total: 19]

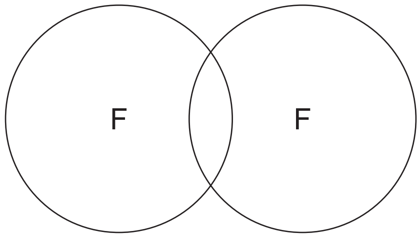
2 Complete the table to:

- deduce the number of protons, electrons and neutrons in the boron atom and chloride ion shown
- identify the atom or ion represented by the final row.

formula	number of protons	number of electrons	number of neutrons
$^{11}_{5}B$		5	
$^{35}_{17}Cl^{-}$	17		
	24	21	30

[Total: 5]

(e) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of fluorine, F_2.
Show the outer electrons only.



[1]

(f) The melting points and boiling points of fluorine and sodium fluoride are shown.

	melting point / °C	boiling point / °C
fluorine	–220	–188
sodium fluoride	993	1695

(i) Deduce the physical state of fluorine at –195 °C. Use the data in the table to explain your answer.

physical state ....

explanation ....

.... [2]

(ii) Explain, in terms of structure and bonding, why sodium fluoride has a much higher melting point than fluorine.

Your answer should refer to the:
- types of particle held together by the forces of attraction
- types of forces of attraction between particles
- relative strength of the forces of attraction.

....

....

....

.... [3]

[Total: 18]

3 Atoms contain protons, neutrons and electrons.

(a) Complete the table to show the relative mass and the relative charge of a proton, a neutron and an electron.

	relative mass	relative charge
proton		
neutron		
electron	$\frac{1}{1840}$	

[3]

(b) The table shows the number of protons, neutrons and electrons in some atoms and ions.

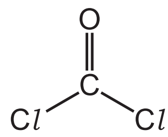
Complete the table.

atom or ion	number of protons	number of neutrons	number of electrons
$^{32}_{16}S$			
$^{39}_{19}K^+$			
	35	44	36

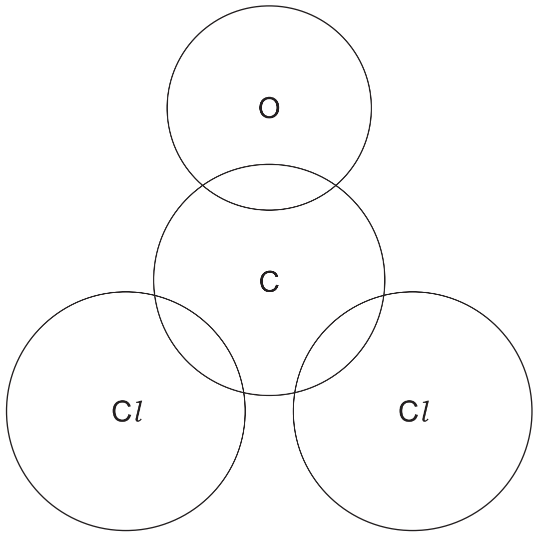
[5]

[Total: 8]

(e) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of $COCl_2$.



Show outer electrons only.

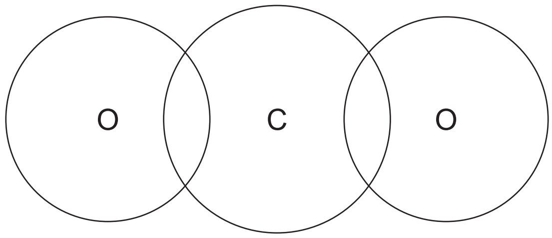


[3]

[Total: 17]

(f) Lead(II) oxide and carbon dioxide are oxides of Group IV elements.

(i) Complete the diagram to show the electron arrangement in one molecule of CO_2.
Show only the outer electrons.



[2]

(ii) The melting points of lead(II) oxide and carbon dioxide are shown.

	melting point / °C
lead(II) oxide	886
carbon dioxide	–56

Use your knowledge of structure and bonding to explain why lead(II) oxide has a much higher melting point than carbon dioxide.

Your answer should refer to:

- the types of particles involved
- the relative strength of the forces of attraction between the particles.

....

....

....

.... [3]

3 The Periodic Table is a method of classifying elements.

(a) Identify the element which is in Group VI and Period 4.

.... [1]

(b) Calcium is in Group II and chlorine is in Group VII of the Periodic Table.

Explain, in terms of number of outer shell electrons and electron transfer, how calcium atoms and chlorine atoms form ions. Give the formulae of the ions formed.

....

....

....

....

....

....

.... [5]

(c) Group V chlorides are covalent molecules. The boiling points of some Group V chlorides are shown.

chloride	boiling point / °C
NCl_3	71
PCl_3	
$AsCl_3$	130
$SbCl_3$	283

(i) Suggest the approximate boiling point of PCl_3.

.... [1]

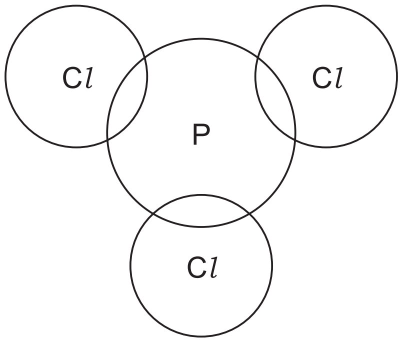
(ii) Explain the trend in boiling points in terms of attractive forces between particles.

....

.... [2]

(iii) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of PCl_3.

Show outer electrons only.



[3]

(d) PCl_3 reacts with chlorine, Cl_2, to form PCl_5. This reaction is exothermic and reaches an equilibrium.

$$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$$

(i) Describe **two** features of an equilibrium.

....

....

.... [2]

(ii) State the effect, if any, on the position of this equilibrium when the following changes are made.
Explain your answers.

temperature is increased ....

....

pressure is increased ....

....
[4]

(iii) Explain, in terms of particles, what happens to the rate of the forward reaction when the reaction mixture is heated.

....

....

....

....

.... [3]

2 Magnesium is a metal.

(a) Name and describe the bonding in magnesium.

name ....

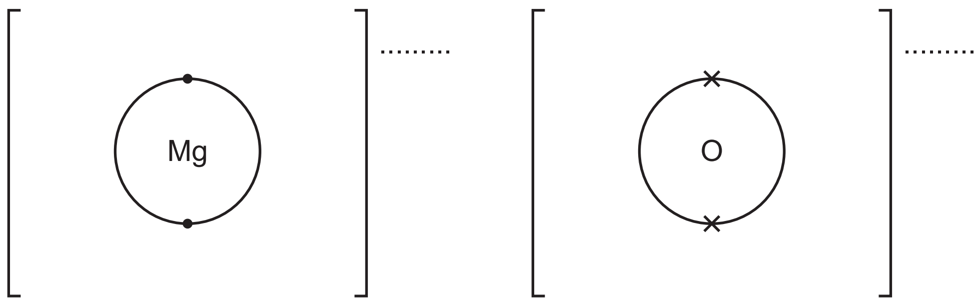
description of bonding ....

....

.... [4]

(b) Magnesium oxide, MgO, is formed when magnesium burns in oxygen.

(i) Complete the dot-and-cross diagram to show the electron arrangement of the ions in magnesium oxide.
The inner shells have been drawn.
Give the charges on the ions.



[3]

(ii) Write the chemical equation for the reaction that occurs when magnesium burns in oxygen.

.... [2]

(c) Magnesium oxide also forms when magnesium nitrate, $Mg(NO_3)_2$, is heated strongly. This is an endothermic reaction.

(i) Write the chemical equation for this reaction.

.... [2]

(ii) What type of reaction is this?

.... [1]

(iii) Name **two** other compounds of magnesium that form magnesium oxide when heated.

....

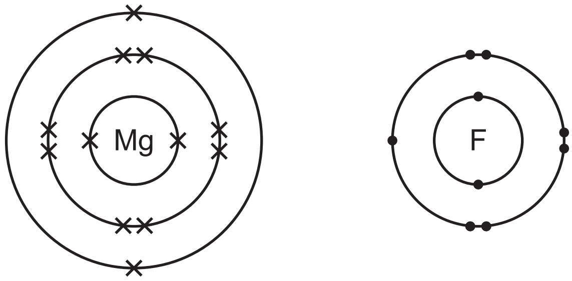
.... [2]

[Total: 14]

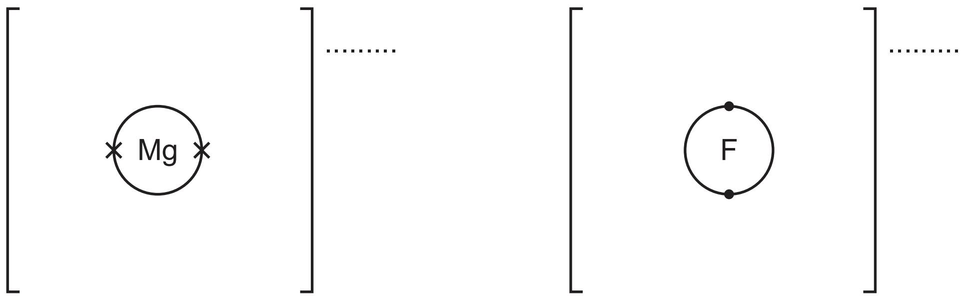
2 Fluorine forms both ionic and covalent compounds.

(a) Magnesium reacts with fluorine to form the ionic compound magnesium fluoride.

The electronic structures of an atom of magnesium and an atom of fluorine are shown.



(i) Complete the dot-and-cross diagrams to show the electronic structures of one magnesium ion and one fluoride ion. Show the charges on the ions.



[3]

(ii) What is the formula of magnesium fluoride?

.... [1]

(iii) Magnesium fluoride does **not** conduct electricity when it is solid.

What can be done to solid magnesium fluoride to make it conduct electricity?

In your answer explain why magnesium fluoride conducts electricity when this change is made.

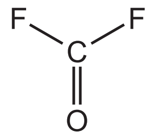
....

....

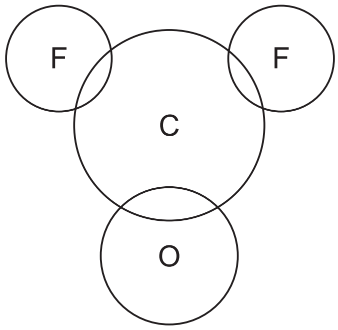
....

.... [2]

(b) Carbonyl fluoride, COF_2, is a covalent compound. The structure of a molecule of COF_2 is shown.



Complete the dot-and-cross diagram to show the electron arrangement in a molecule of carbonyl fluoride. Show outer shell electrons only.



[3]

(c) The melting points of magnesium fluoride and carbonyl fluoride are shown.

	melting point / °C
magnesium fluoride	1263
carbonyl fluoride	–111

(i) Explain, using your knowledge of structure and bonding, why magnesium fluoride has a high melting point.

....

....

....

.... [2]

(ii) Explain, using your knowledge of structure and bonding, why carbonyl fluoride has a low melting point.

....

....

....

.... [2]

[Total: 13]

3 Chlorine is in Group VII of the Periodic Table.

(a) Two isotopes of chlorine are chlorine-35 and chlorine-37.

(i) State why these two isotopes of chlorine have the same chemical properties.

....

....

.... [2]

(ii) Complete the table to show the number of electrons, neutrons and protons in each atom and ion.

	number of electrons	number of neutrons	number of protons
${}^{35}_{17}Cl$			
${}^{37}_{17}Cl^{-}$			

[3]

(b) **(i)** Chlorine reacts with aqueous sodium bromide.

The equation for the reaction is shown.

$$Cl_2 + 2NaBr \rightarrow 2NaCl + Br_2$$

State the type of reaction shown.

.... [1]

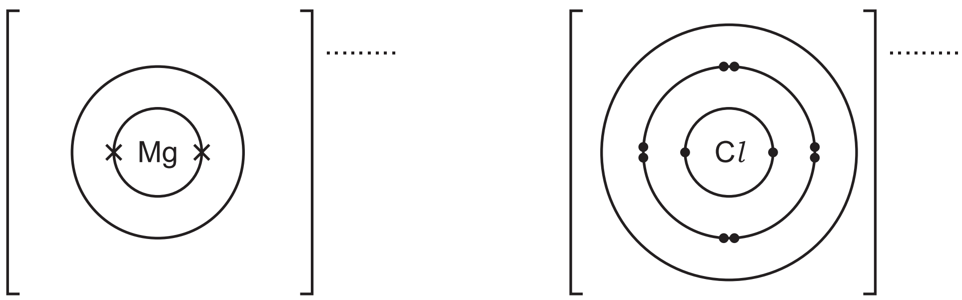
(ii) Why is there **no** reaction between iodine and aqueous sodium bromide?

.... [1]

(c) Magnesium reacts with chlorine to form magnesium chloride.

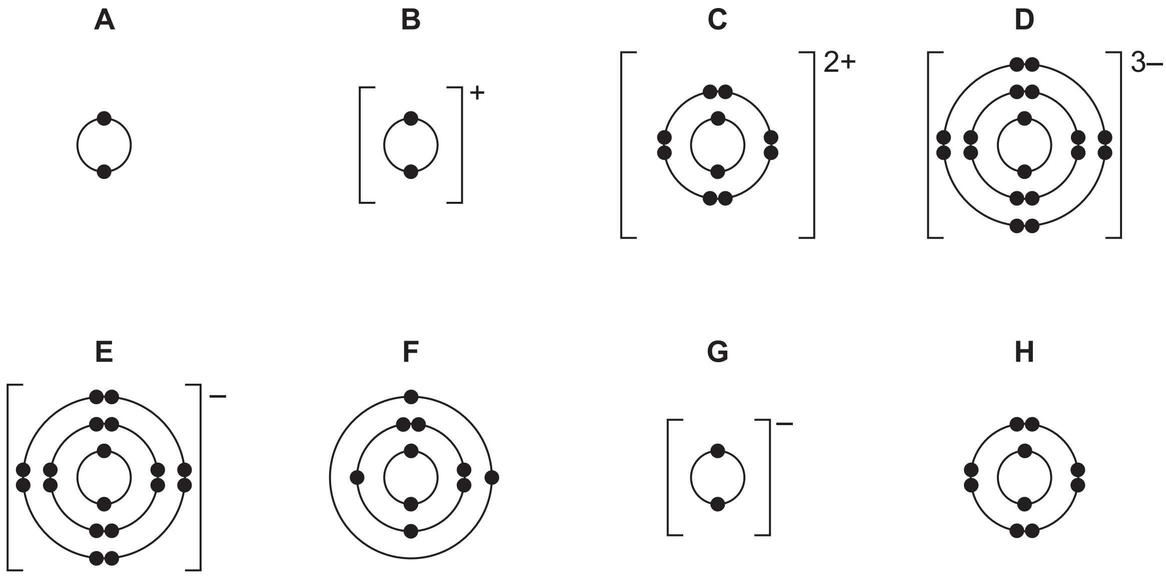
Complete the dot-and-cross diagram to show the electron arrangement of the ions in magnesium chloride. Give the charges on the ions.

The inner shells have been completed.



[3]

1 The electronic structures of some atoms and ions are shown.



(a) Write the letters, **A**, **B**, **C**, **D**, **E**, **F**, **G** or **H**, of the electronic structures which show:

(i) atoms of two different noble gases .... and ....
[2]

(ii) an ion of a Group I element .... [1]

(iii) an ion of a Group V element .... [1]

(iv) a pair of ions that could form a compound with the formula XY_2. .... and .... [1]

(b) State which electronic structure, **A**, **B**, **C**, **D**, **E**, **F**, **G** or **H**, is incorrect.

Explain why.

incorrect electronic structure ....

explanation ....

....
[2]

(c) State how many protons are found in the nucleus of ion **C**. .... [1]

(d) Use the Periodic Table to deduce:

(i) the chemical symbol for ion **G** .... [1]

(ii) the element which forms an ion with a 3+ charge and the same electronic structure as **H**.

.... [1]

[Total: 10]

2 The table gives information about five particles, **A**, **B**, **C**, **D** and **E**.

particle	number of electrons	number of neutrons	number of protons
A	10	13	11
B	18	20	18
C	18	18	18
D	10	12	8
E	10	10	10

(a) State the atomic number of **A**.

.... [1]

(b) State the nucleon number of **B**.

.... [1]

(c) Write the electronic structure of **C**.

.... [1]

(d) Give the letters of all the particles which are:

(i) atoms .... [1]

(ii) positive ions .... [1]

(iii) negative ions .... [1]

(iv) isotopes of each other. .... [1]

[Total: 7]